

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected Graduation, 2027

East Lansing, MI

- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 – Present

East Lansing, MI

- Replaced multi-day manual onboarding with one automated provisioning flow (organization membership plus Slack workspace invites) so new contributors receive repository and channel access in under 30 seconds, cutting setup time an estimated 97% across more than 100 members.
- Retired paper attendance by issuing short-lived QR tokens cached at the edge with S3 and CloudFront, validating check-in in real time and sustaining 150+ scans per event without redundant authentication round-trips.
- Hardened how the club ships software with GitHub Actions coverage gates above 90%, branch protection on production merges, and review standards that reduced escaped defects by an estimated 65%.

PROJECTS

Dadei — Ambient AI Assistant | *Python, FastAPI, PostgreSQL, pgvector, Redis, RQ, Gemini API, faster-whisper, pyannote-audio, PyTorch, Docker*

May 2025 – Nov. 2025

dadei.app

- Built an ambient assistant that closes the loop from passive, multi-device audio to structured outcomes: after each conversation segment, Gemini 2.5 Pro proposes grounded episodic memories and calendar-style actions with evidence quotes tied back to a numbered transcript manifest, reducing brittle wake-word command UX.
- Ran local speech understanding on CPU using **faster-whisper** (**large-v3-turbo**, int8) for word-timed transcripts, **pyannote/speaker-diarization-3.1** for segmentation, and an online centroid + capped prototype bank with cosine gates to stabilize who-spoke-what across sessions; diarization-attributed turns improved speaker-consistency score by **[TBD]**% on held-out clips.
- Implemented hybrid RAG over **pgvector** in Postgres: Gemini **gemini-embedding-001** vectors (768-d, L2-normalized) with IVFFlat cosine ANN, then application-level semantic-lexical fusion, recency decay, and optional reciprocal-rank fusion before prompting—raising top-k useful context hits from **[TBD]**% → **[TBD]**% on an internal replay suite.
- Shipped the full inference and persistence stack under Docker with Alembic migrations (including the **vector** extension), optional Redis/RQ workers, and CI that boots a real **pgvector** service for pytest—the same pipeline now backs **100+** concurrently registered personal networks with bounded duplicate delivery after Redis dedupe claims.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, TensorFlow, NumPy*

Feb. 2025

devpost.com/software/wizviz

- Won first place in the Interactive Media track at SpartaHack X with a gesture-driven wizard duel where live body pose becomes spell casting, blocking, and ability triggers against an on-device AI opponent—no game engine, no cloud round-trips.
- Mapped full-body pose to gameplay by extracting 33 skeletal landmarks each frame, classifying them into discrete actions, and tightening the inference loop until the experience held 60 FPS with sub-20ms input lag.
- Projected live 3D pose into a hand-built 2D physics model with collision response so motion remained stable during play at 60 FPS.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Golang, C++, Bash

ML & AI PyTorch, faster-whisper, pyannote-audio, Google Gemini API, pgvector, TensorFlow, MediaPipe, spaCy

Frameworks FastAPI, Electron, Vite, React, Redis, RQ, Docker

Databases & Infrastructure PostgreSQL, GitHub Actions, AWS (S3, CloudFront)